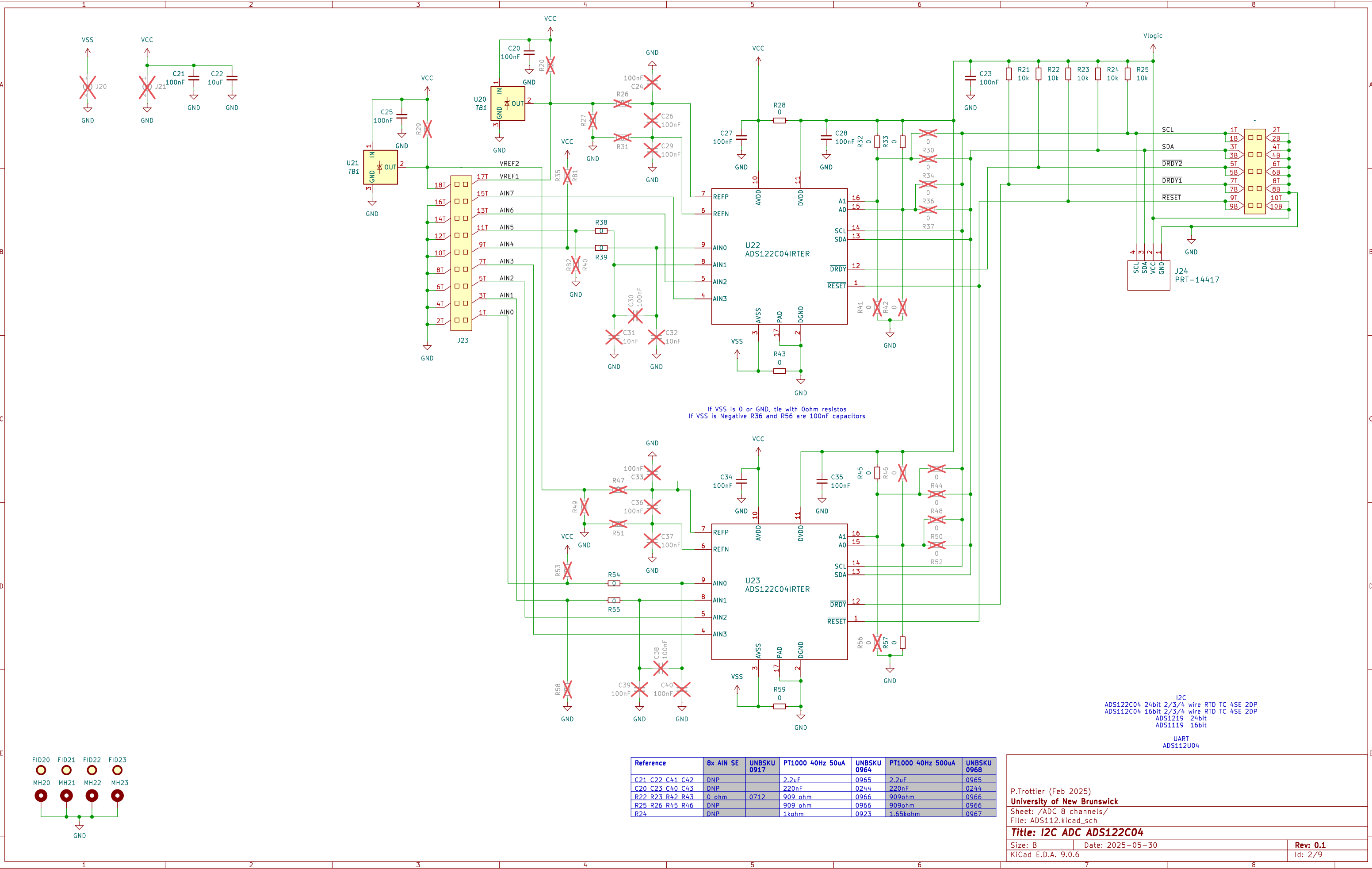




Reference	8x AIN SE	UNBSKU 0917	PT1000 40Hz 50uA	UNBSKU 0964	PT1000 40Hz 500uA	UNBSKU 0968
C21 C22 C41 C42	DNP		2.2uF	0965	2.2uF	0965
C20 C23 C40 C43	DNP		220nF	0244	220nF	0244
R22 R23 R42 R43	0 ohm	0712	909 ohm	0966	909ohm	0966
R25 R26 R45 R46	DNP		909 ohm	0966	909ohm	0966
R24	DNP		1kohm	0923	1.65kohm	0967

I2C
ADS122C04 24bit 2/3/4 wire RTD TC 4SE 2DP
ADS122C04 16bit 2/3/4 wire RTD TC 4SE 2DP
ADS1219 24bit
ADS1119 16bit

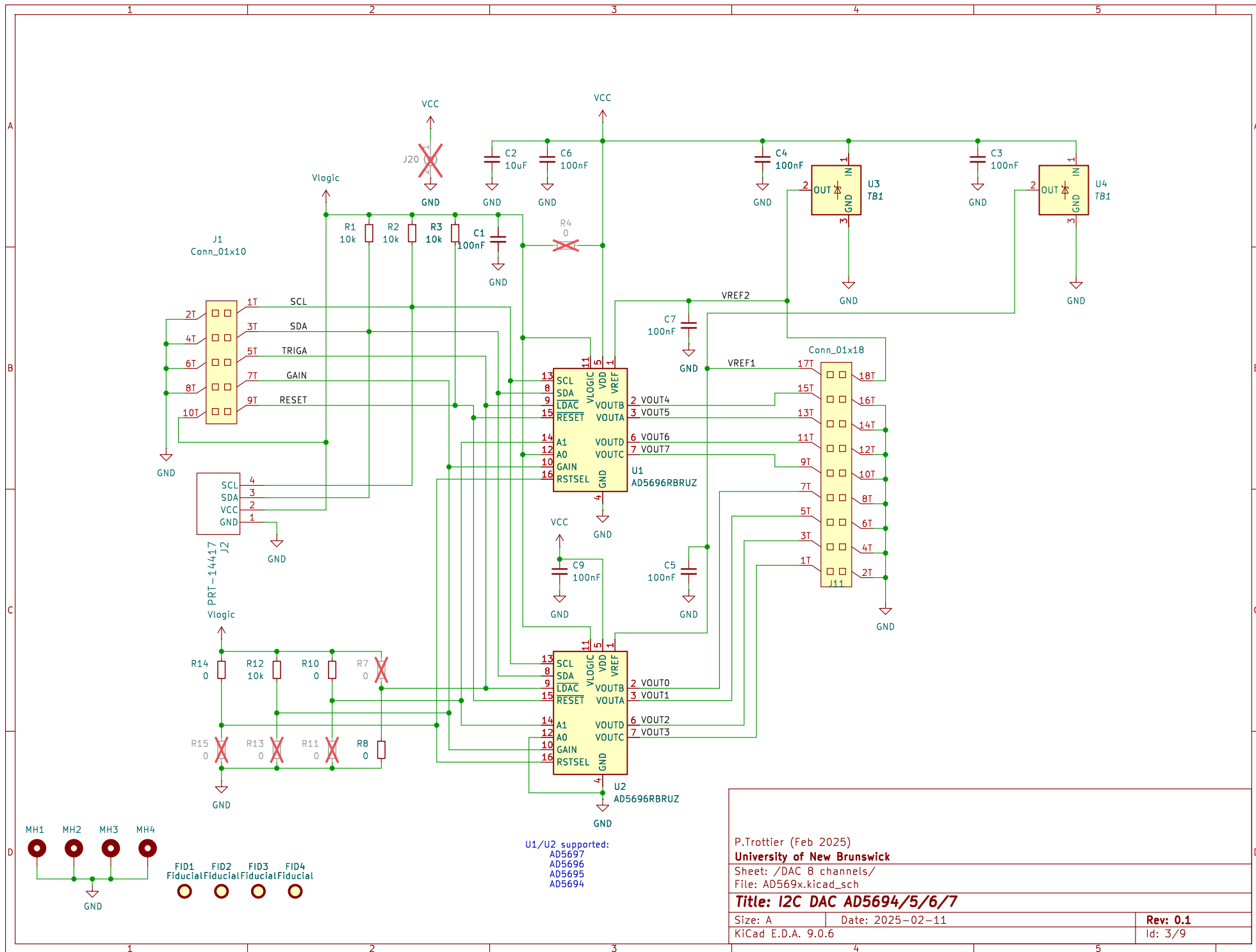
UART
ADS112U04

P.Trottier (Feb 2025)
University of New Brunswick
Sheet: /ADC 8 channels/
File: ADS112.kicad_sch

Title: I2C ADC ADS122C04

Size: B Date: 2025-05-30 **Rev: 0.1**

KiCad E.D.A. 9.0.6 Id: 2/9

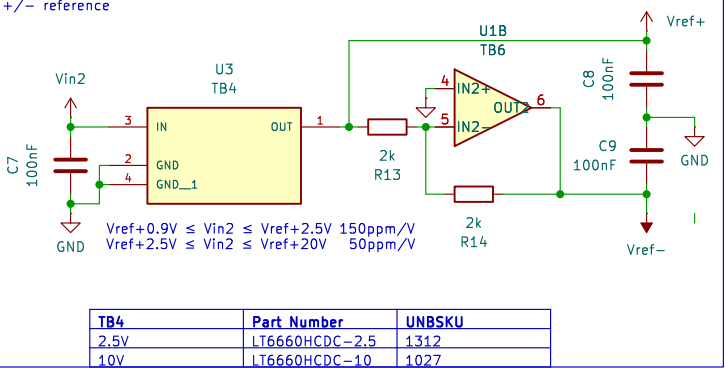


Power supply

TB2	R15	UNBSKU
+2.5V	11.0k	1307
+2.73V	12.9k	1311
+2.76V	13.2k	1310
+3.0V	15.4k	1301
+3.3V	17.8k	0921
+5.0V	32.4k	1298
+11.9V	89.8k	1210
+12.0V	90.9k	1305
+12.3V	93.1k	1304
+13.1V	100k	0710

TB3	R18	UNBSKU
-2.5V	21.0k	1306
-2.73V	22.9k	1309
-2.76V	23.2k	1308
-3.0V	25.5k	1302
-3.3V	28.0k	0810
-5.0V	42.2k	1297
-12V	100k	0710
-13.1V	110k	1303

+/- reference



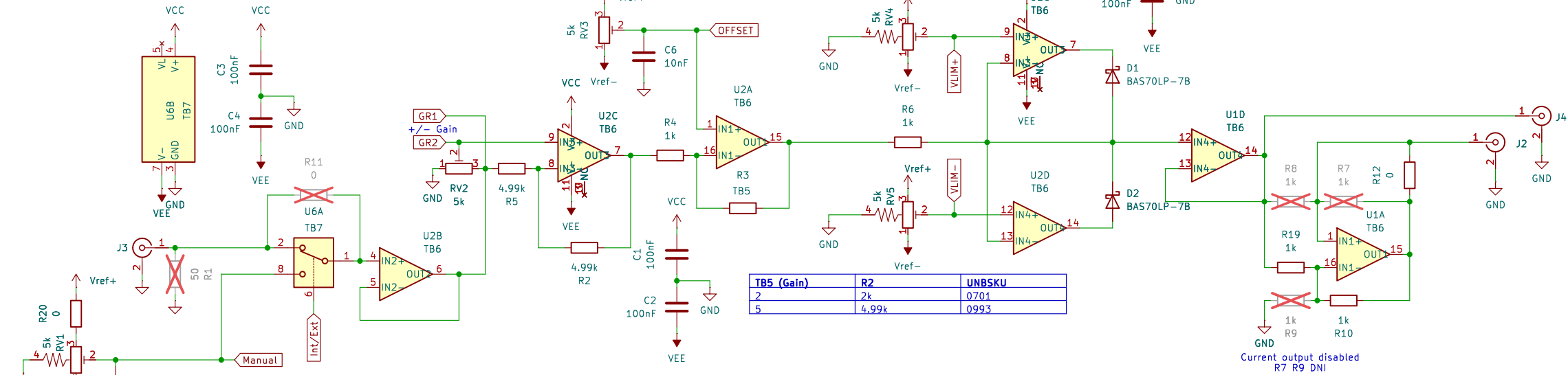
TB4	Part Number	UNBSKU
2.5V	LT6660HDCDC-2.5	1312
10V	LT6660HDCDC-10	1027

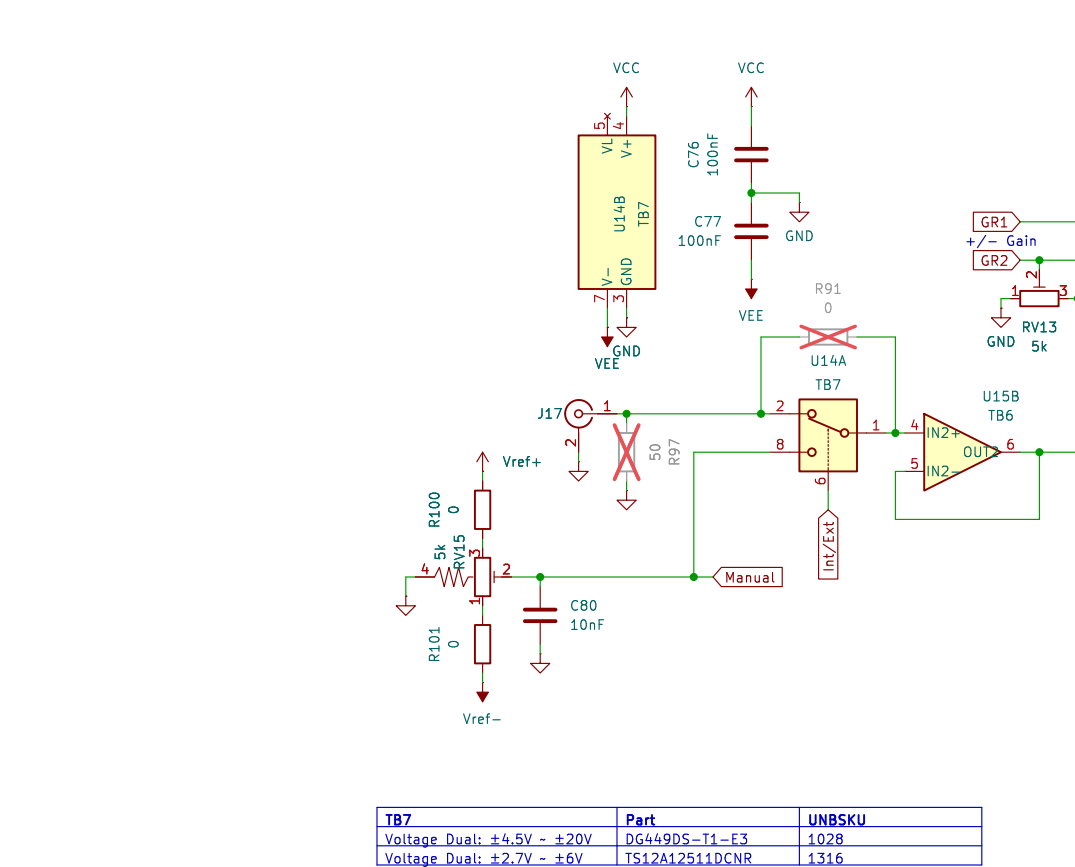
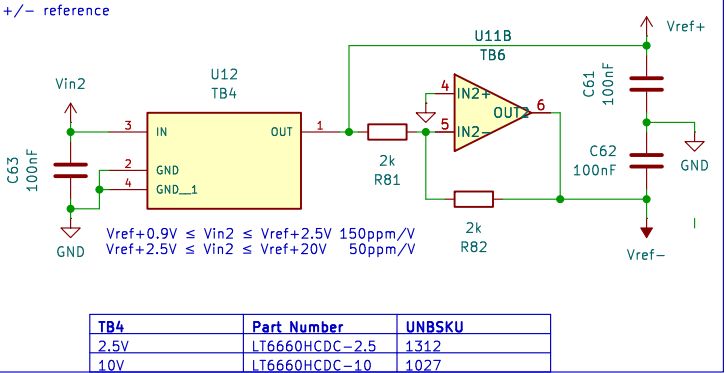
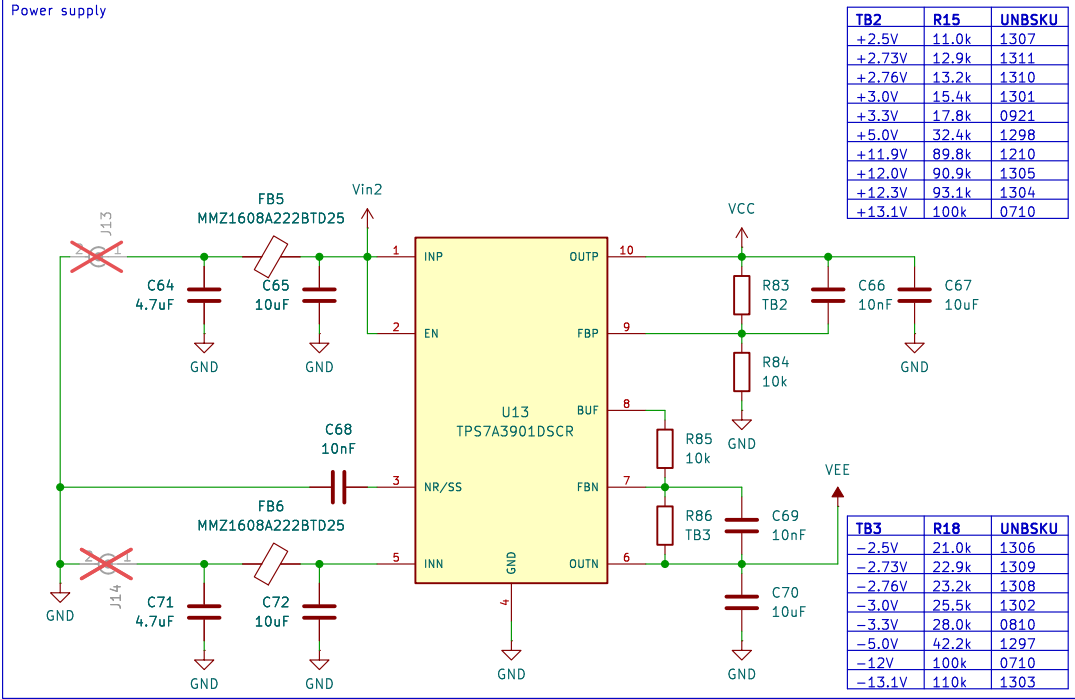
For RV1, RV3, RV4, RV5:
 $V_{set} = V_{ref}(TB4) - (r1/(r1+r2)) * (2 * V_{ref}(TB4))$

For RV2:
 $GainSet = Vin * 1 - (r1/(r1+r2) * 2)$

For all RV1-5 :
 $r1 + r2 > (V(TB2) - V(TB3)) / 0.005A$

For b = 0 : C6 = 0ohm



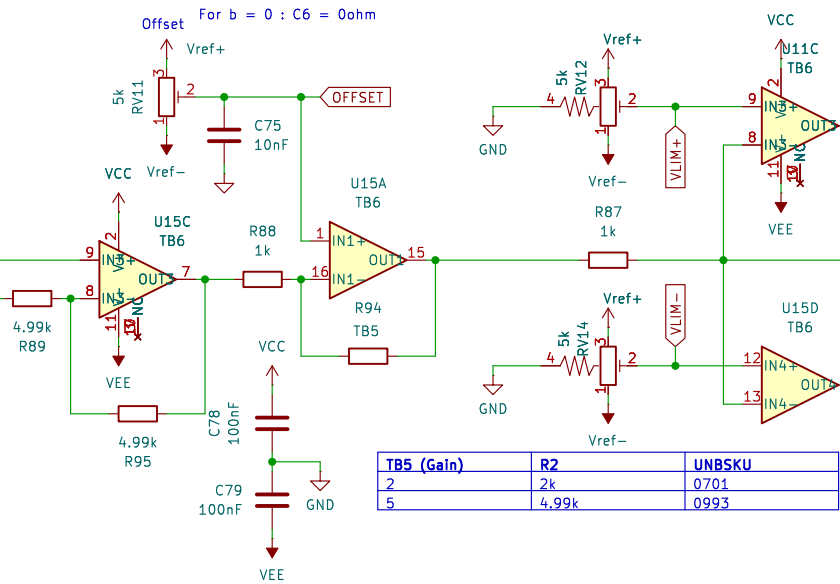


For RV1, RV3, RV4, RV5:
 $V_{set} = V_{ref}(TB4) - (r1/(r1+r2)) * (2 * V_{ref}(TB4))$

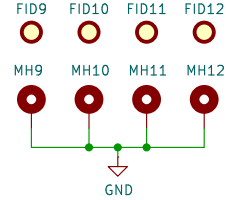
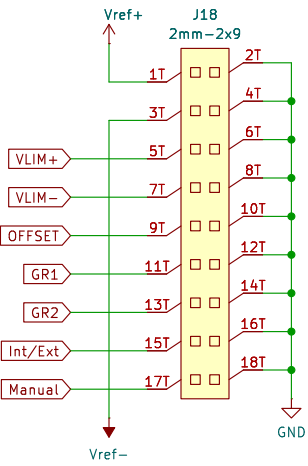
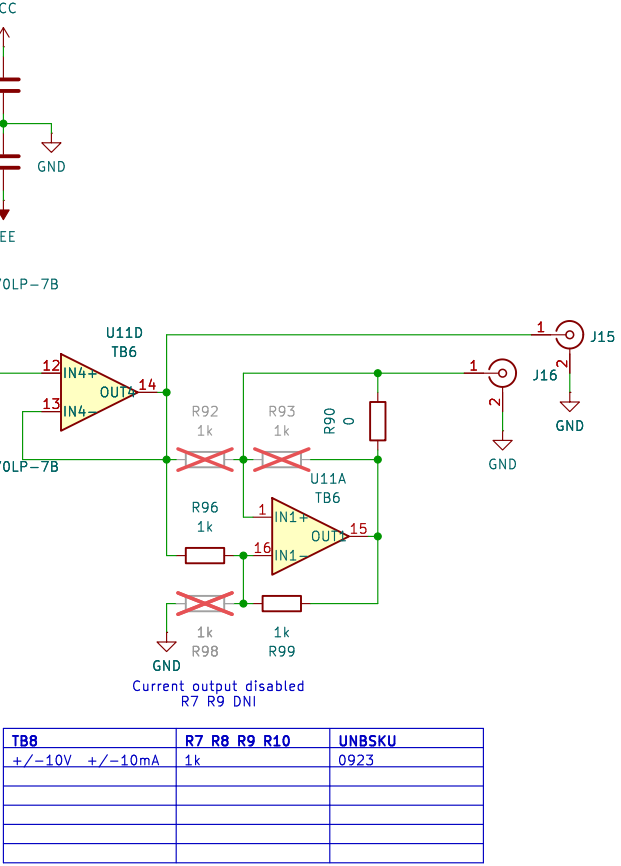
For RV2:
 $GainSet = Vin * 1 - (r1/(r1+r2) * 2)$

For all RV1-5 :
 $r1 + r2 > (V(TB2) - V(TB3)) / 0.005A$

For b = 0 : C6 = 0ohm



TB6	Part	UNBSKU
BW: 1.3MHz Rail: ±15V Output: 20mA Offset: 2mV	LM2902Q4T	1029
BW: 20MHz Rail: ±2.75V Output: 55mA Offset: 200µV	TSV774IQ5T	1314
BW: 400kHz Rail: ±2.75V Output: 18mA Offset: 1µV	OA4ZHA33Q	1315

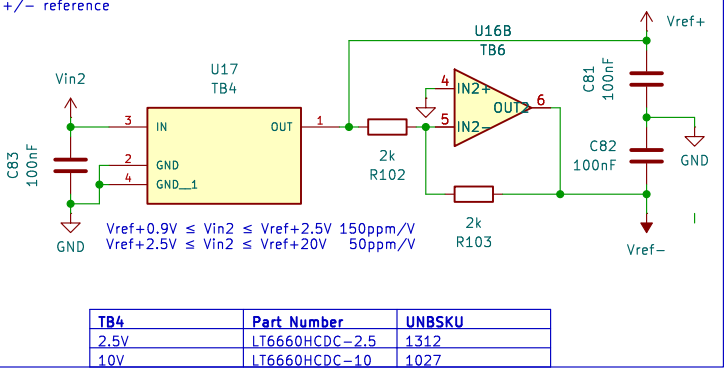


Power supply

TB2	R15	UNBSKU
+2.5V	11.0k	1307
+2.73V	12.9k	1311
+2.76V	13.2k	1310
+3.0V	15.4k	1301
+3.3V	17.8k	0921
+5.0V	32.4k	1298
+11.9V	89.8k	1210
+12.0V	90.9k	1305
+12.3V	93.1k	1304
+13.1V	100k	0710

TB3	R18	UNBSKU
-2.5V	21.0k	1306
-2.73V	22.9k	1309
-2.76V	23.2k	1308
-3.0V	25.5k	1302
-3.3V	28.0k	0810
-5.0V	42.2k	1297
-12V	100k	0710
-13.1V	110k	1303

+/- reference



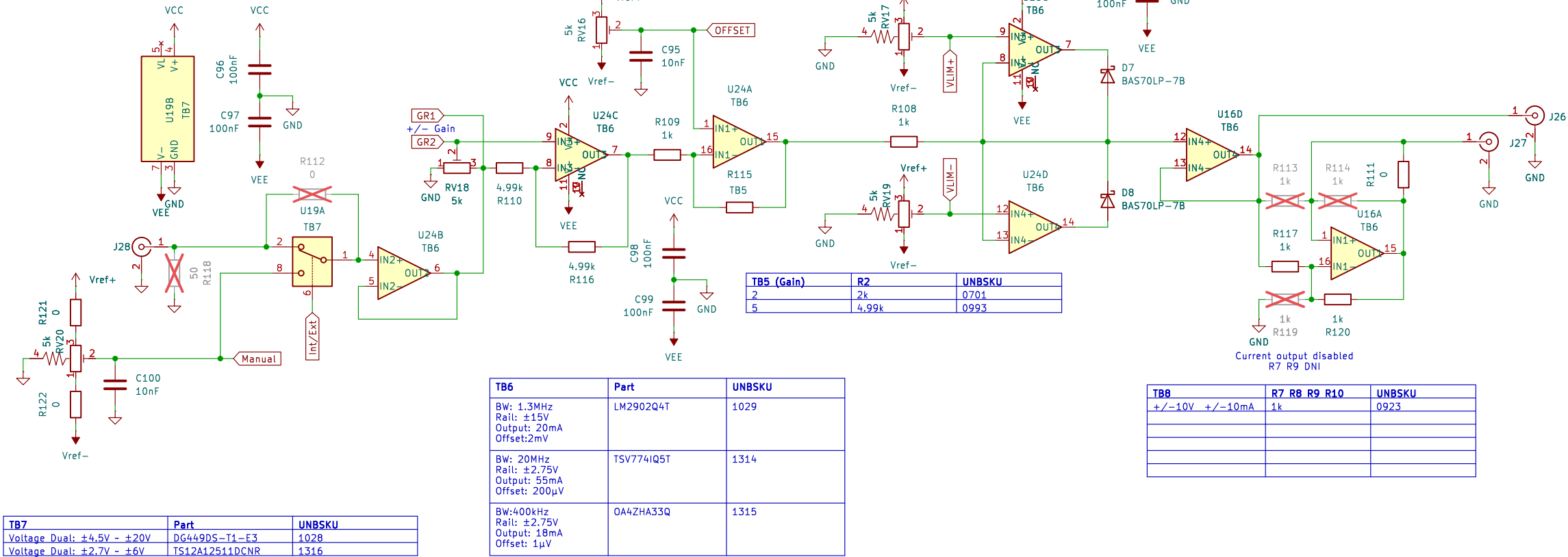
TB4	Part Number	UNBSKU
2.5V	LT6660HDCDC-2.5	1312
10V	LT6660HDCDC-10	1027

For RV1,RV3,RV4,RV5:
 $V_{set} = V_{ref}(TB4) - (r1/(r1+r2)) * (2 * V_{ref}(TB4))$

For RV2:
 $GainSet = V_{in} * 1 - (r1/(r1+r2) * 2)$

For all RV1-5 :
 $r1 + r2 > (V(TB2) - V(TB3)) / 0.005A$

For b = 0 : C6 = 0ohm

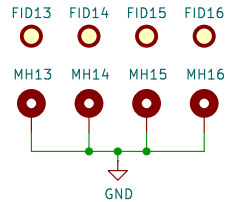
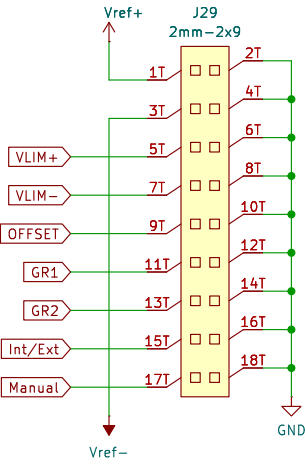


TB7	Part	UNBSKU
Voltage_Dual: ±4.5V ~ ±20V	DG449DS-T1-E3	1028
Voltage_Dual: ±2.7V ~ ±6V	TS12A12511DCNR	1316

TB6	Part	UNBSKU
BW: 1.3MHz Rail: ±15V Output: 20mA Offset: 2mV	LM2902Q4T	1029
BW: 20MHz Rail: ±2.75V Output: 55mA Offset: 200µV	TSV774IQ5T	1314
BW: 400kHz Rail: ±2.75V Output: 18mA Offset: 1µV	OA4ZHA33Q	1315

TB5 (Gain)	R2	UNBSKU
2	2k	0701
5	4.99k	0993

TB8	R7 R8 R9 R10	UNBSKU
+/-10V +/-10mA	1k	0923

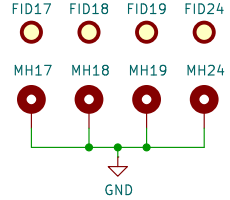
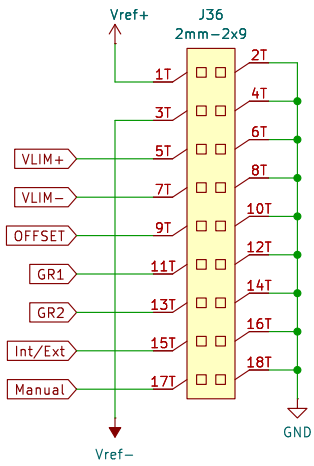
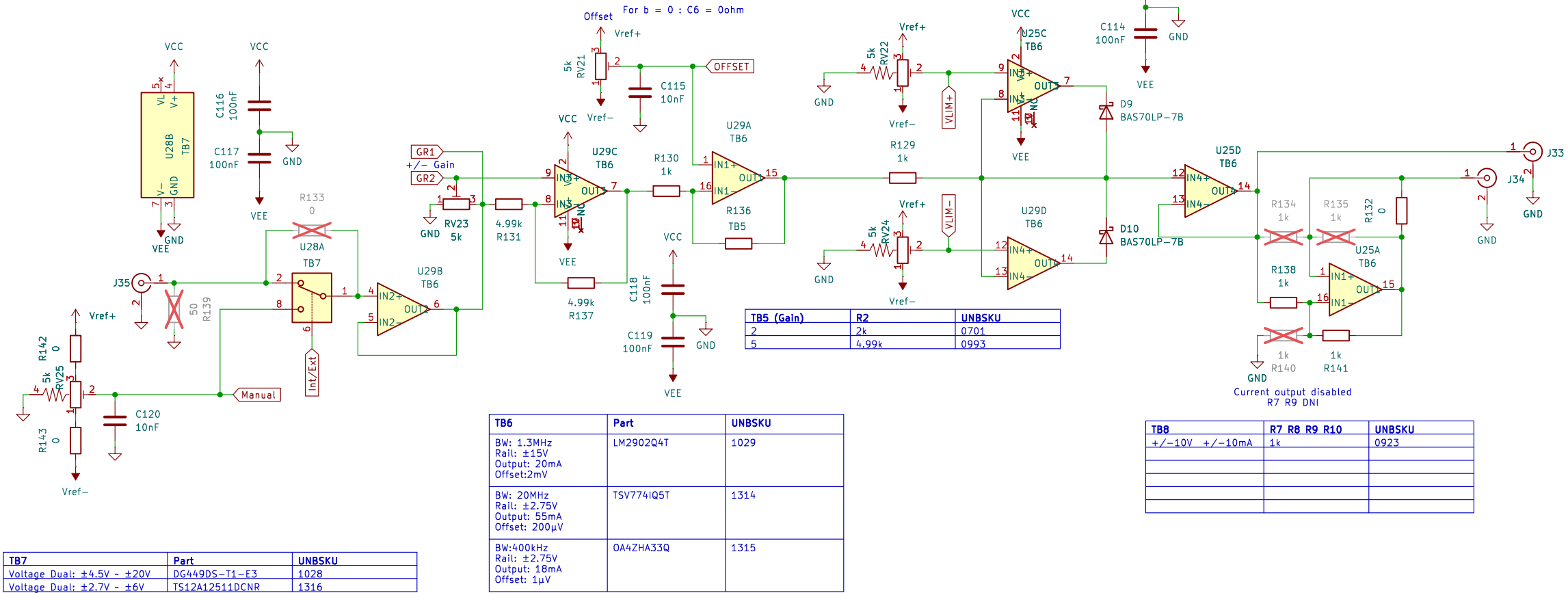
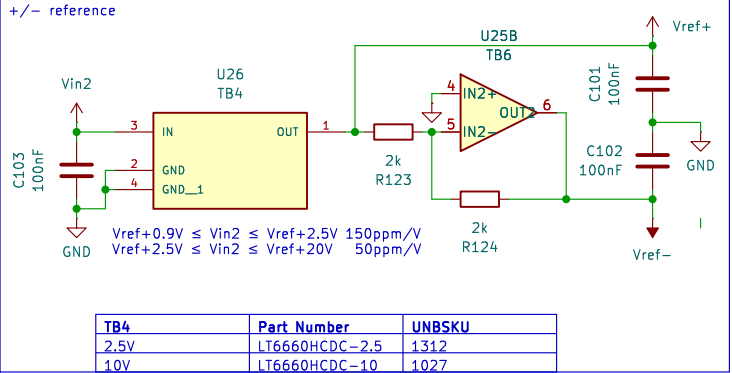
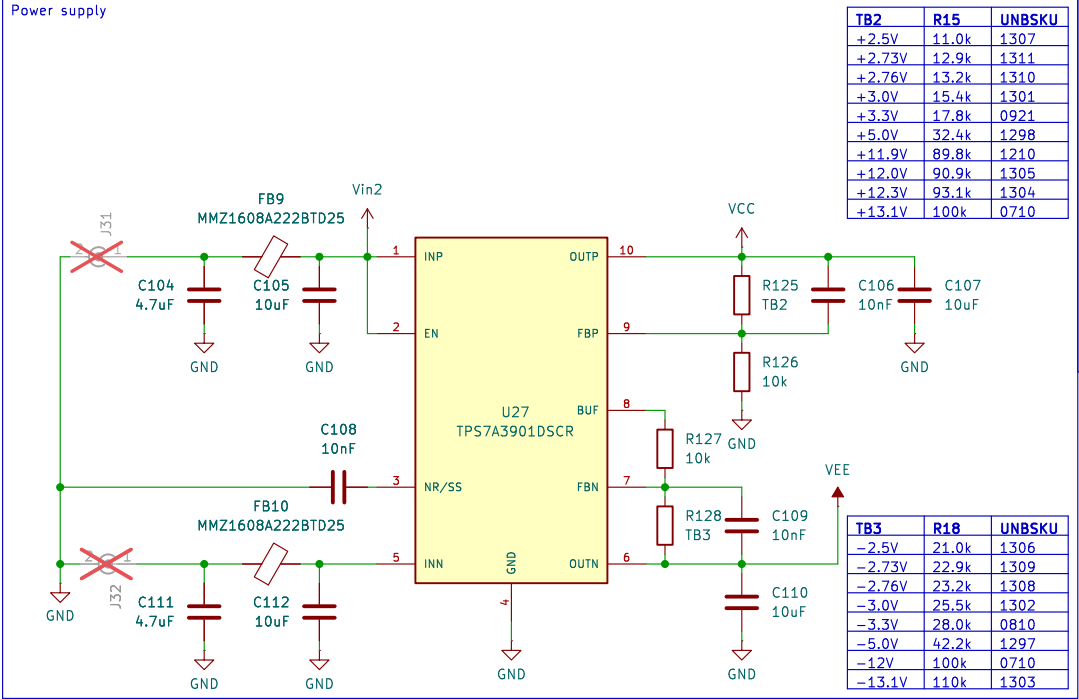


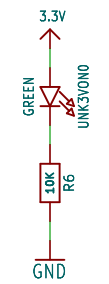
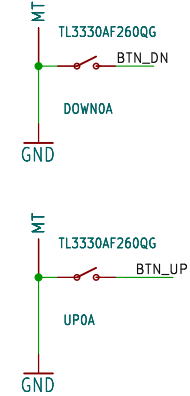
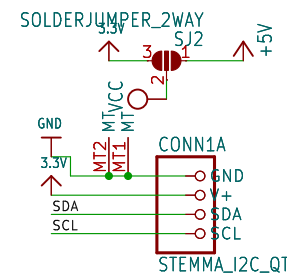
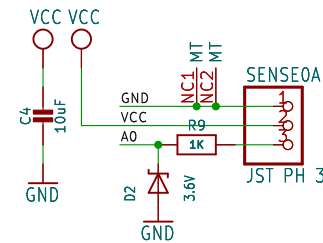
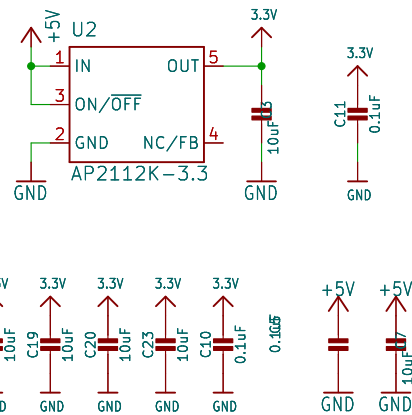
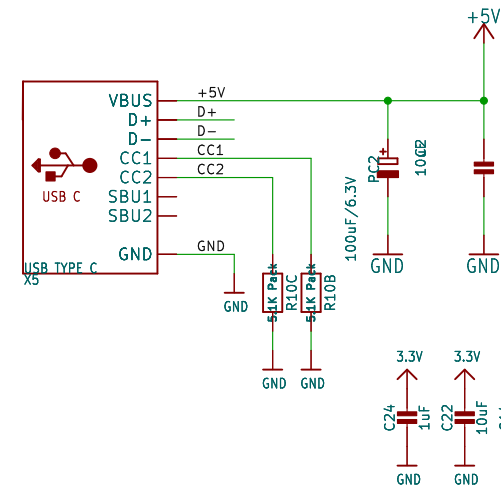
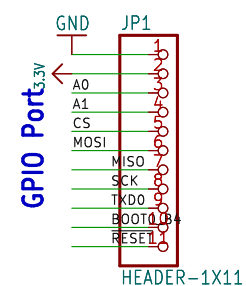
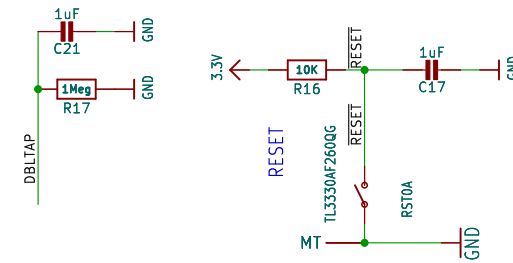
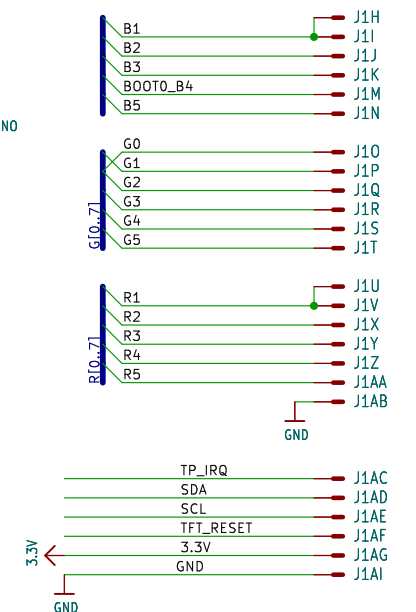
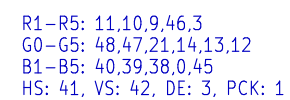
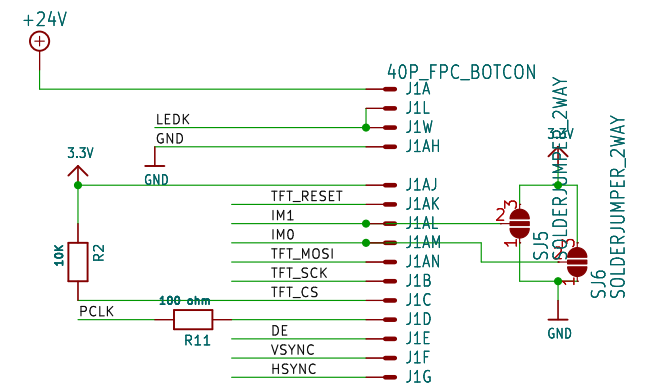
P.Trottier (2025)
University of New Brunswick
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File: CondCtrl.kicad_sch

Title: Condition and Manual Ctrl

Size: B Date: 2025-11-18
KiCad E.D.A. 9.0.6

Rev: 0.1
Id: 7/9



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